

Internship Write-Up

Testing and Analysis of
Automotive Embedded Control Systems

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1DS22EC078 | VIII Sem B

Department of ECE | DSCE

Carried out at HUEGLI TECH ENGINEERING Pvt. Ltd.

January 2026 -- May 2026

SEE -- May / June 2026 | 22INT82

INTERNSHIP WRITE-UP SPEECH

OPENING

Good morning / Good afternoon, respected faculty, Dr. Thenmozhi Ma'am, and my fellow students.

I am Harish Kumar R B, USN 1DS22EC078, from the eighth semester, Section B. I am here to present my internship write-up on "Testing and Analysis of Automotive Embedded Control Systems", carried out at Huegli Tech Engineering Private Limited, Bangalore, from January 1st to May 12th, 2026.

SECTION 1 -- Details Regarding the Internship

Company

I completed my internship at Huegli Tech Engineering Pvt. Ltd., located at Number 46, BGS Enclave, Anepalya, B.M. Kaval, Bangalore -- 560060. Huegli Tech is the Indian subsidiary of Huegli Tech AG, a Switzerland-based company specialising in industrial electronics and automation systems for diesel and gas engine applications.

The company develops, tests, and repairs products such as the HT-SG-100 Speed Switch Governor, the RM Mixer Controller, and various industrial embedded control units. Their work spans PCB design, firmware validation, CAN Bus communication, actuator control, and quality verification of industrial electronic systems.

The internship was offline, and I obtained this opportunity independently -- not through the college placement cell. No stipend was provided, and it was not an in-house project.

Domain

My internship domain was Industrial Embedded Electronics and Automotive Control Systems. The core technologies involved were:

- * Modbus communication over USB HID
- * PIC microcontroller-based embedded control
- * CAN Bus industrial communication
- * Signal conditioning and actuator control
- * PCB testing and hardware fault analysis

Department

I was placed in the Testing and Repair Department. This department is responsible for testing, troubleshooting, repairing, and validating industrial electronic systems and PCB boards before they are deployed in real industrial environments. The department handles quality inspection, hardware verification, communication testing, performance analysis, and PCB repair activities including soldering, desoldering, and component replacement.

SECTION 2 -- Project Done During the Internship

Aim

The aim of my internship project was to understand, test, and validate the HT-SG-100 Speed Switch Governor and RM Mixer Controller systems -- two key industrial products developed by Huegli Tech -- by performing systematic hardware and software testing procedures, identifying faults, and verifying correct operation of all subsystems.

Block Diagram Overview

HT-SG-100 Speed Switch Governor

A diesel or gas engine drives a flywheel. A magnetic pickup sensor (MPU) mounted near the flywheel detects the passing of teeth at a 1 to 2 millimetre air gap. This generates an AC signal whose frequency is proportional to the engine RPM. The HT-SG-100 controller receives this AC signal, processes it using a PIC microcontroller, converts it to RPM, displays it on a seven-segment display, and uses it to control an actuator -- which physically adjusts the fuel supply to maintain stable engine speed.

The system also includes relay driver circuits for overspeed and overcurrent protection, CAN Bus and USB communication interfaces, an EEPROM for parameter storage, and an RTC for event logging.

RM Mixer Controller

The RM Mixer Controller is a companion system used to regulate gas-air mixture flow in gas engine applications. It processes analogue input signals in ranges of 0-5V, 0-10V, 0-20mA, and 4-20mA, and drives a stepper motor to control the mixer valve position.

Methodology

Step 1 -- Visual Inspection:

Each PCB board was visually examined for damaged components, cold solder joints, burnt components, and physical defects before powering on.

Step 2 -- Power Supply Testing:

The 24V input, regulated 5V, and 3.3V output rails were verified using a digital multimeter to ensure the power section was stable.

Step 3 -- Startup Verification:

The controller booting sequence and seven-segment display initialisation were observed during power-on.

Step 4 -- Signal Testing:

RPM sensing was tested using a pickup signal simulator. Analogue inputs, feedback signals, gain, idle, and droop inputs were verified using external signal sources.

Step 5 -- Functional Testing:

Actuator forward and reverse movement, relay outputs, LED indications, buzzer functionality, and keypad operation were tested under different operating conditions.

Step 6 -- Communication Testing:

USB communication was verified using the PC configuration software (PCLab). CAN Bus communication was verified using the PCAN Viewer tool.

Step 7 -- Protection Testing:

Overspeed protection was tested by applying higher-than-threshold RPM signals. Overcurrent protection was tested by simulating fault conditions at the actuator terminals.

Step 8 -- EEPROM Validation:

Parameter storage and retrieval functions were verified by writing values, power cycling the controller, and confirming the values were retained.

Step 9 -- PCB Repair:

Faulty boards identified during testing were repaired through soldering correction, component replacement, and in some cases, epoxy removal and board restoration.

Step 10 -- Final Validation:

A full functional test was carried out to confirm stable operation of all subsystems before a board was cleared for deployment.

Tools Used

Hardware Tools:

- * Digital Multimeter -- for voltage, continuity, and signal verification
- * Soldering Station -- for PCB repair and component replacement
- * Desoldering Tools -- for removing faulty components
- * USB-to-UART Converter -- for firmware programming
- * CAN Communication Device (PCAN) -- for CAN Bus testing
- * RPM / Pickup Signal Simulator -- for RPM signal generation
- * Regulated Power Supply Unit -- for providing DC supply during testing

Software Tools:

- * EAGLE -- PCB schematic and layout understanding
- * MPLAB -- PIC microcontroller programming and debugging
- * PCAN Software -- CAN Bus communication monitoring
- * PCLab -- parameter configuration and controller telemetry monitoring
- * Renesas Flash Programmer -- firmware flashing and bootloader operations

Results and Conclusion

The testing and validation activities yielded the following results:

- * All HT-SG-100 Speed Switch Governor units tested showed correct RPM sensing, stable actuator control, functional relay outputs, and working overspeed and overcurrent protection.
- * USB and CAN Bus communication were verified successfully on all units.
- * RM Mixer systems demonstrated proper analogue signal conversion, stable stepper motor operation in both clockwise and counterclockwise directions, and correct feedback output across all configured operating modes.
- * PCB troubleshooting and repair activities identified and resolved faults caused by soldering defects, damaged passive components, and signal path failures.
- * Firmware programming, EEPROM storage, and controller boot sequences functioned correctly on all repaired units.

In conclusion, the internship provided me with direct hands-on exposure to industrial-grade embedded electronic systems, professional testing methodologies, and real-world fault diagnosis techniques used in a production environment.

SECTION 3 -- Learning Outcomes

Technical Outcomes

1. PCB Troubleshooting and Repair:

Learned to identify faults at the component level and perform precision soldering, desoldering, and component replacement on industrial PCBs.

2. Industrial Embedded Systems:

Gained a practical understanding of PIC microcontroller-based control systems, EEPROM storage, real-time PID control loops, and firmware validation.

3. CAN Bus Communication:

Tested and verified CAN Bus communication using professional tools, understanding how industrial controllers exchange data on a shared bus.

4. Modbus over USB HID:

Developed a deep understanding of how Modbus RTU protocol operates over a USB HID link, including register read and write operations using function codes FC03, FC04, and FC16.

5. Signal Analysis and Actuator Testing:

Learned to use signal simulators, multimeters, and test equipment to verify analogue and digital signals at various test points in industrial circuits.

Non-Technical Outcomes

1. Teamwork and Communication:

Working alongside experienced engineers taught me how to collaborate effectively and communicate technical findings clearly.

2. Industrial Work Exposure:

I experienced the discipline, pace, and responsibility of a professional industrial work environment.

3. Time Management:

Managing multiple testing tasks simultaneously within production timelines improved my ability to prioritise and work efficiently.

4. Problem-Solving:

Diagnosing faults in complex electronic systems without full documentation trained me to think methodically and systematically.

5. Confidence:

Handling industrial systems gave me the confidence to work independently with professional-grade electronic equipment.

CLOSING

To summarise, my internship at Huegli Tech Engineering gave me four months of intensive, practical experience in industrial embedded electronics -- from testing and repairing speed governors and gas-air mixer controllers, to working with CAN Bus, USB communication, and Modbus protocol at the byte level.

This experience has significantly strengthened my technical foundation and prepared me for a professional career in embedded systems and industrial electronics.

Thank you.

Estimated delivery time: 8 -- 10 minutes at a comfortable pace.